

Base Paper vs NeuroShift

Abstract, Conclusion, Hardware and Software comparison | Base: Chamberland et al., IEEE EMBC 2024

Base paper: safer EMG GESTURE recognition for prosthesis using vision. NeuroShift: safer HOME INTENTION control using vision + sparse EMG with Act/Abstain and FAR metric.

1. Abstract comparison

1.1 Base paper abstract (summary)

Title: Visual Scene Understanding for Enhanced EMG Gesture Recognition

A multimodal approach combining electromyography (EMG) and computer vision for robust real-time gesture recognition. Context-aware framework for myoelectric prosthesis control: EMG hand gesture recognition is augmented by visual detection of objects, mitigating false movements. SDCNN EMG predictions are supported by a tailored YOLO model. Pilot experiment shows improved robustness while keeping user command.

- Focus: gesture recognition robustness
- Domain: myoelectric prosthesis
- Vision role: reduce false gesture detections
- Output: gesture class (grip type)

1.2 NeuroShift abstract (draft for our project)

NeuroShift is a multimodal AI framework for assistive smart-home control. The camera detects appliances and which one the user is looking at (gaze/head direction). Sparse EMG then confirms whether they truly want to toggle it. Looking alone never actuates. The system Acts or Abstains, targeting lower False Actuation Rate (FAR). Existing Indian wired appliances are controlled via ESP32+MQTT relay hub. Phase 1: head-gaze + EMG demo. Phase 2: eye-gaze, learned fusion, priors, XAI, IEEE evaluation.

- Focus: gaze-select + EMG-confirm + safe actuation
- Domain: assistive smart home / independent living
- Vision role: detect appliances + select by gaze
- EMG role: confirm intent only (not gesture classes)
- Output: device action OR abstain

1.3 Abstract side-by-side

Point	Base paper	NeuroShift
Problem	False EMG gestures in prosthesis control	False/unsafe home device actuations
Users	Upper-limb prosthesis users	Motor-impaired users with residual EMG
Predicts	Hand gesture / grip	Home device intention or abstain
Vision job	Validate/filter grips	Build affordance candidates
EMG job	Classify gesture	Confirm volition/effort
Safety idea	Fewer false movements	Act/Abstain + FAR
Setting	Lab prosthesis scenarios	Indian wired home appliances

2. Conclusion comparison

2.1 Base paper conclusion (summary)

Pilot work shows vision-based object detection can provide context for EMG gesture recognition while preserving user agency. SDCNN confidence helps when EMG is ambiguous. Vision fusion corrected many false predictions in real time. Challenges remain: crowded scenes, occlusion, unknown objects, and better EMG/vision confidence fusion. Future work: improve YOLO for amputee needs and explore temporal confidence fusion.

- Proved: EMG + vision reduces gesture errors
- Kept: user still chooses grip
- Open issues: multi-object clutter, occlusion, unknown objects

2.2 NeuroShift conclusion (draft)

NeuroShift extends scene-aware EMG-vision fusion from prosthetic gesture rescue to assistive smart-home intention inference. By treating sparse EMG as volitional confirmation and vision as an affordance proposer, the system can Act on a target appliance or Abstain under ambiguity, targeting lower False Actuation Rate in multi-object domestic scenes. Integration via MQTT relays/smart plugs makes the approach compatible with conventional Indian wired homes. Phase 1 delivers a working prototype and pilot dataset; Phase 2 provides learned fusion, personal priors, decision-level explanations, and IEEE-oriented evaluation against EMG-only, vision-only, and Chamberland-style baselines.

- Proves: intention + abstention can reduce false home actuations
- Keeps: user agency via EMG confirmation
- Addresses base-paper gap: multi-object ambiguity via Abstain

2.3 Conclusion side-by-side

Point	Base paper	NeuroShift
Main finding	Vision context improves gesture robustness	Affordance + EMG + abstain improves safe home control
Primary metric spirit	Gesture errors fixed / accuracy	FAR + intention F1
User agency	User still selects grip	User confirms volition; system may refuse action
Future work they left	Crowded scenes, better fusion	We take crowded scenes as core Act/Abstain problem
Deployment	Prosthesis control concept	ESP32 hub for normal wired homes

3. Hardware comparison

Item	Base paper (used)	NeuroShift (we will use)
EMG sensor	EMaGer 64-channel HD-EMG @ 1 kHz (custom lab sensor)	MyoWare 2.0 sparse EMG (commercial)
EMG channels	64 (high-density array)	1 (expandable to 2 later)
Camera	EMEET C960 webcam, head-mounted, 640x480	USB webcam / laptop webcam (ambient room view)
Compute	PC + GTX 1080 GPU (reported for YOLO)	Laptop GPU/CPU (student setup)
Actuation target	Prosthesis gesture/grip output (control interface)	Lamp / fan / plug via relays or smart plugs
IoT / home layer	Not for home appliances	ESP32 + MQTT + relay module / smart plug
Indian wired homes	Not addressed	Yes - retrofit relay interface + manual override
Wearable form	HD-EMG bracelet (EMaGer)	MyoWare + optional 3D-printed armband
Demo appliances	Mug, apple, bottle, phone (grasp scenarios)	Real on/off devices (lamp, fan, plug)
Cost level	Research HD-EMG (expensive/custom)	Approx. Rs 10k-15k student kit

Why we change hardware

- EMaGer is custom 64-ch research hardware - not practical for our timeline/budget.
- Our novelty is intention + Act/Abstain + FAR, not HD-EMG sensor design.
- MyoWare is enough for volition confirmation in assistive home control.
- ESP32/MQTT is required because we control real home appliances.

4. Software / algorithm comparison

Module	Base paper	NeuroShift
EMG model	SDCNN (Siamese Deep CNN) + centroid classifier	RMS/MAV volition score; Phase2 small TCN/1D-CNN
Gesture/intent classes	6 grips: open, tripod, power, thumbs-up, pinch, point	Device actions: lamp/fan/plug + ABSTAIN
Object detection	Custom YOLOv8-small (28 classes)	YOLO11/YOLOv8/RT-DETR (home devices)
Selector	MediaPipe Hands near object	Phase1 head/face gaze; Phase2 eye-gaze
Fusion method	OLF enables object-related grips	Gaze selects + EMG confirms + Act/Abstain
If unsure	Mostly still gesture pathway / defaults	Explicit ABSTAIN (looking != wanting)
Backend / UI	Real-time processing pipeline	Python + FastAPI + React dashboard + logger
IoT software	Not applicable	MQTT (Mosquitto) + ESP32 firmware
Database	Not central	SQLite (P1) / PostgreSQL (P2) trial logs
Primary metric	Gesture errors reduced	False Actuation Rate (FAR) + intention F1
XAI	Limited (confidence scores)	Phase 2 decision-level explanations
Personalization	No	Phase 2 temporal/habit prior

5. Items list - Base paper used

Hardware

- EMaGer HD-EMG bracelet (64 channels, 1 kHz)
- EMEET C960 head-mounted webcam
- PC with GPU (GTX 1080 mentioned for YOLO inference)

Software / models

- SDCNN (Siamese CNN) for EMG gesture recognition
- YOLOv8-small custom-trained object detector
- MediaPipe Hands
- Object Likelihood Filtering (OLF) fusion logic
- Training data context from COCO + OpenImages (28 classes)

6. Items list - NeuroShift will use

Hardware (Phase 1)

- MyoWare 2.0 muscle sensor + electrode pads
- ESP32 DevKit (1-2)
- Relay module and/or MQTT smart plug
- USB/laptop webcam
- Laptop for AI inference
- Table lamp + desk fan (demo) compatible with Indian wired homes
- Optional: 3D-printed armband + hub enclosure

Software (Phase 1)

- Python, PyTorch, OpenCV
- YOLOv8 or RT-DETR + MediaPipe/RTMPose
- Rule-based Act/Abstain intention policy
- FastAPI + React dashboard
- MQTT (Mosquitto) + ESP32 firmware
- SQLite/CSV trial logger

Software (Phase 2 add)

- Learned fusion model (MLP / small cross-attention)
- Calibrated abstention for FAR
- Personal temporal prior
- Decision-level XAI + caregiver views
- PostgreSQL + full evaluation scripts

7. One-page takeaway for guide

Layer	They did	We do
Abstract aim	Better prosthesis gestures with vision	Safer home intentions with Act/Abstain
Conclusion aim	Vision reduces EMG gesture errors	Vision+EMG+abstain reduces false home actuations
EMG hardware	Custom 64-ch EMaGer	MyoWare sparse EMG
Vision software	YOLO + MediaPipe + OLF	YOLO/RT-DETR + pose + affordance policy
Output	Grip class	Device action or Abstain
Home integration	No	ESP32 + MQTT relays for wired appliances

We keep the base paper's multimodal idea (EMG + vision), change the problem (home intention), change the safety policy (Act/Abstain), change the metric (FAR), and change the hardware to a low-cost MyoWare + ESP32 stack suitable for

Indian homes and college implementation.